

Yen-Hsiang (Robert) Huang

✉ yhuang861@gatech.edu ☎ +1(470)923-6469

Educations

Ph.D. in Electrical and Computer Engineering GPA: 4.0/4.0 Georgia Institute of Technology (Gatech), Atlanta, GA, USA	Sept. 2022 - Present
B.S. in Electrical Engineering GPA: 4.06/4.30 National Taiwan University (NTU), Taipei, Taiwan	Sept. 2017 - July 2022

Research Experience

Graduate Research Assistant

GTCAD Lab., Georgia Institute of Technology, Advisor: Prof. Sung Kyu Lim	Sept. 2022 - Present
3D IC Optimization through 3D-VIA Reduction & Inter-Die Cell Relocation	Fall 2024 - Present
○ Timing optimization through reduced critical path snaking and inter-die cell relocation.	
Semiconductor Research Corporation (SRC) – JUMP 2.0 CHIMES	Fall 2023 - Spring 2024
○ Bridge the PPA gap between legacy node 3D ICs and advanced nodes 2D ICs ○ Reduced WNS of legacy node 3D ICs by 50-93% compared to SOTA, achieving 87.9-97.7% of advanced node 2D performance	
Die Bonding Bumps and Pads legalization for 3D ICs	Sept. 2022 - May 2023
○ Developed a 3D via legalization stage with a refinement step to legalize 3D vias with minimize runtime impact ○ Our flow is capable of eliminating all 3D via overlaps when via utilization is less than 40%	

Individual Study

EDA Lab., National Taiwan University, Advisor: Prof. Yao-Wen Chang	Apr. 2020 - Sept. 2021
Independent Study on Physical Design	
○ Participated in physical-design-related global contests three times in different teams, and get into the top 5 two times (ISPD contest 2021, ICCAD contest 2021)	
ALCom Lab., National Taiwan University, Advisor: Prof. Jie-Hong Jiang	Sept. 2020 - June 2021
Independent Study on Reversible circuit Synthesis	
○ Implemented a reversible circuit synthesizer for controlled-NOT (CNOT) gates	

Work Experience

R&D Engineer

Synopsys Inc.	Synopsys Headquarters May 2025 - Dec. 2025
○ Investigated AI-based signal-integrity (SI) analysis for 2.5D ICs, using graph neural networks (GNNs) to predict final-stage SI performance from early-stage design data. ○ Independently implemented two provided prototypes in C++ and merged the resulting code into the main branch.	

Software Engineer R&D Intern

Synopsys Inc. - Concurrent Clock-and-Data (CCD) Team	Hsinchu, Taiwan July 2021 - Aug. 2021
○ Enhanced the flip-flop merging algorithm in Synopsys IC Compiler II (ICC II) by considering the flip-flop's timing (slack) ○ Improved the timing quality in 13 real industrial benchmarks: 5.14%/3.47% in worst/total negative slack	

Software Engineer R&D Intern

Cadence Inc. - Conformal Team	Hsinchu, Taiwan June 2020 - Sept. 2020
○ Provided electronic change order (ECO) test cases in a 5-person team for Cadence Conformal's robustness test ○ Generated 2000+, 5 kinds of cases in which Conformal acts unusually and proposed 3 improved flows to fix	

Publication

- **Yen-Hsiang Huang** and Sung Kyu Lim, "CLK-3D: RC-Calibrated Differentiable Clock-Tree Optimization for 3D ICs," IEEE/ACM International Conference on Computer-Aided Design, 2026.
- **Yen-Hsiang Huang** and Sung Kyu Lim, "Snake-3D: Differentiable Learning for Cross-Tier Logic Path Snaking Optimization in 3D ICs," IEEE/ACM International Conference on Computer-Aided Design (ICCAD), 2025. [PDF]
- **Yen-Hsiang Huang**, Sai Pentapati, Anthony Agnesina, Moritz Brunion, and Sung Kyu Lim, "On Legalization of Die Bonding Bumps and Pads for 3D ICs," IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2024.
- Sai Pentapati, Anthony Agnesina, Moritz Brunion, **Yen-Hsiang Huang**, and Sung Kyu Lim, "On Legalization of Die Bonding Bumps and Pads for 3D ICs", ACM International Symposium on Physical Design (ISPD), 2023.

Honors and Awards

Honorable Mention Award in the ACM ISPD contest 2021

March 2021

- 4th place worldwide in the contest.

TSMC scholarship for undergraduate student

Sept. 2020 - Jan. 2021

- Awarded to approximately 20 undergraduate students in Taiwan every semester.

The Dean's List Award

Spring 2020

- Ranked 1st, with 4.30/4.30 GPA in the semester.